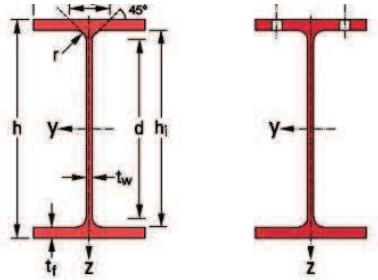
Avrupa kesit
I-profiller

IPE

IPE 100 - 600
Euronorm 19-57
IPE A 100 - 600
IPE O 180 - 600
IPE 750



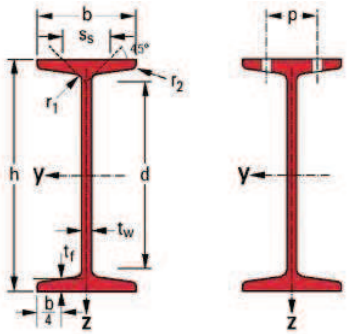
1	2	3	4	5			13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28

KESİT	ÖLÇÜLER						KESİT ALANI	TASARIM ÖLÇÜLERİ					YÜZEY ALANI		KESİT	STATİK DEĞERLER												
																KUVVETLİ EKSEN y-y					ZAYIF EKSEN z-z							
	G	h	b	t_w	t_f	r	A	h_i	d	Ø	pmin	Pmax	A_L	A_G		G	I_y	$W_{el.y}$	$W_{pl.y}$	i_y	A_{vz}	I_z	$W_{el.z}$	$W_{pl.z}$	i_z	s_s	I_t	$I_w x 10^{-3}$
	kg/m	mm	mm	mm	mm	mm	cm^2	mm	mm		mm	mm	m^2/m	m^2/t		kg/m	cm^4	cm^3	cm^3	cm	cm^2	cm^4	cm^3	cm^3	cm	mm	cm^4	cm^6
IPE A 100	6.9	98	55	3.6	4.7	7	8.78	88.6	74.6	-	-	-	0.397	57.57	IPE A 100	6.9	141.2	28.81	32.98	4.01	4.44	13.12	4.77	7.54	1.22	21.20	0.77	0.28
IPE 100	8.1	100	55	4.1	5.7	7	10.32	88.6	74.6	-	-	-	0.400	49.33	IPE 100	8.1	171.0	34.20	39.41	4.07	5.08	15.92	5.79	9.15	1.24	23.70	1.20	0.35
IPE A 120	8.7	117.6	64	3.8	5.1	7	11.03	107.4	93.4	-	-	-	0.472	54.47	IPE A 120	8.7	257.4	43.77	49.87	4.83	5.41	22.39	7.00	10.98	1.42	22.20	1.04	0.71
IPE 120	10.4	120	64	4.4	6.3	7	13.21	107.4	93.4	-	-	-	0.475	45.82	IPE 120	10.4	317.8	52.96	60.73	4.90	6.31	27.67	8.65	13.58	1.45	25.20	1.74	0.89
IPE A 140	10.5	137.4	73	3.8	5.6	7	13.39	126.2	112.2	-	-	-	0.547	52.05	IPE A 140	10.5	434.9	63.30	71.60	5.70	6.21	36.42	9.98	15.52	1.65	23.20	1.36	1.58
IPE 140	12.9	140	73	4.7	6.9	7	16.43	126.2	112.2	-	-	-	0.551	42.70	IPE 140	12.9	541.2	77.32	88.34	5.74	7.64	44.92	12.31	19.25	1.65	26.70	2.45	1.98
IPE A 160	12.7	157	82	4.0	5.9	9	16.18	145.2	127.2	-	-	-	0.619	48.70	IPE A 160	12.7	689.3	87.81	99.09	6.53	7.80	54.43	13.27	20.70	1.83	26.34	1.96	3.09
IPE 160	15.8	160	82	5.0	7.4	9	20.09	145.2	127.2	-	-	-	0.623	39.47	IPE 160	15.8	869.3	108.7	123.9	6.58	9.66	68.31	16.66	26.10	1.84	30.34	3.60	3.96
IPE A 180	15.4	177	91	4.3	6.5	9	19.58	164.0	146.0	M10	48	48	0.694	45.15	IPE A 180	15.4	1063	120.1	135.3	7.37	9.20	81.89	18.00	27.96	2.05	27.84	2.70	5.93
IPE 180	18.8	180	91	5.3	8.0	9	23.95	164.0	146.0	M10	48	48	0.698	37.13	IPE 180	18.8	1317	146.3	166.4	7.42	11.25	100.9	22.16	34.60	2.05	31.84	4.79	7.43
IPE O 180	21.3	182	92	6.0	9.0	9	27.10	164.0	146.0	M10	50	50	0.705	33.12	IPE O 180	21.3	1505	165.4	189.1	7.45	12.70	117.3	25.50	39.91	2.08	34.54	6.76	8.74
IPE A 200	18.4	197	100	4.5	7.0	12	23.47	183.0	159.0	M10	54	58	0.764	41.49	IPE A 200	18.4	1591	161.6	181.7	8.23	11.47	117.2	23.43	36.54	2.23	32.56	4.11	10.53
IPE 200	22.4	200	100	5.6	8.5	12	28.48	183.0	159.0	M10	54	58	0.768	34.36	IPE 200	22.4	1943	194.3	220.6	8.26	14.00	142.4	28.47	44.61	2.24	36.66	6.98	12.99
IPE O 200	25.1	202	102	6.2	9.5	12	31.96	183.0	159.0	M10	56	60	0.779	31.05	IPE O 200	25.1	2211	218.9	249.4	8.32	15.45	168.9	33.11	51.89	2.30	39.26	9.45	15.57
IPE A 220	22.2	217	110	5.0	7.7	12	28.26	201.6	177.6	M12	60	62	0.843	38.02	IPE A 220	22.2	2317	213.5	240.2	9.05	13.55	171.4	31.17	48.49	2.46	34.46	5.69	18.71
IPE 220	26.2	220	110	5.9	9.2	12	33.37	201.6	177.6	M12	60	62	0.848	32.36	IPE 220	26.2	2772	252.0	285.4	9.11	15.88	204.9	37.25	58.11	2.48	38.36	9.07	22.67
IPE O 220	29.4	222	112	6.6	10.2	12	37.39	201.6	177.6	M10	58	66	0.858	29.24	IPE O 220	29.4	3134	282.3	321.1	9.16	17.66	239.8	42.83	66.91	2.53	41.06	12.27	26.79
IPE A 240	26.2	237	120	5.2	8.3	15	33.31	220.4	190.4	M12	64	68	0.918	35.10	IPE A 240	26.2	3290	277.7	311.6	9.94	16.31	240.1	40.02	62.40	2.68	39.37	8.35	31.26
IPE 240	30.7	240	120	6.2	9.8	15	39.12	220.4	190.4	M12	66	68	0.922	30.02	IPE 240	30.7	3892	324.3	366.6	9.97	19.14	283.6	47.27	73.92	2.69	43.37	12.88	37.39
IPE O 240	34.3	242	122	7.0	10.8	15	43.71	220.4	190.4	M12	66	70	0.932	27.17	IPE O 240	34.3	4369	361.1	410.3	10.00	21.36	328.5	53.86	84.40	2.74	46.17	17.18	43.68
IPE A 270	30.7	267	135	5.5	8.7	15	39.15	249.6	219.6	M16	70	72	1.037	33.75	IPE A 270	30.7	4917	368.3	412.5	11.21	18.75	358.0	53.03	82.34	3.02	40.47	10.30	59.51
IPE 270	36.1	270	135	6.6	10.2	15	45.95	249.6	219.6	M16	72	72	1.041	28.86	IPE 270	36.1	5790	428.9	484.0	11.23	22.14	419.9	62.20	96.95	3.02	44.57	15.94	70.58
IPE O 270	42.3	274	136	7.5	12.2	15	53.84	249.6	219.6	M16	72	72	1.051	24.88	IPE O 270	42.3	6947	507.1	574.6	11.36	25.23	513.5	75.51	117.7	3.09	49.47	24.90	87.64

IPE A 300	36.5	297	150	6.1	9.2	15	46.53	278.6	248.6	M16	72	86	1.156	31.65	IPE A 300	36.5	7173	483.1	541.8	12.42	22.25	519.0	69.20	107.3	3.34	42.07	13.43	107.2
IPE 300	42.2	300	150	7.1	10.7	15	53.81	278.6	248.6	M16	72	86	1.160	27.46	IPE 300	42.2	8356	557.1	628.4	12.46	25.68	603.8	80.50	125.2	3.35	46.07	20.12	125.9
IPE O 300	49.3	304	152	8.0	12.7	15	62.83	278.6	248.6	M16	74	88	1.174	23.81	IPE O 300	49.3	9994	657.5	743.8	12.61	29.05	745.7	98.12	152.6	3.45	50.97	31.06	157.7
IPE A 330	43.0	327	160	6.5	10.0	18	54.74	307.0	271.0	M16	78	96	1.250	29.09	IPE A 330	43.0	10230	625.7	701.9	13.67	26.99	685.2	85.64	133.3	3.54	47.59	19.57	171.5
IPE 330	49.1	330	160	7.5	11.5	18	62.61	307.0	271.0	M16	78	96	1.254	25.52	IPE 330	49.1	11770	713.1	804.3	13.71	30.81	788.1	98.52	153.7	3.55	51.59	28.15	199.1
IPE O 330	57.0	334	162	8.5	13.5	18	72.62	307.0	271.0	M16	80	98	1.268	22.24	IPE O 330	57.0	13910	833.0	942.8	13.84	34.88	960.4	118.6	185.0	3.64	56.59	42.15	245.7
IPE A 360	50.2	357.6	170	6.6	11.5	18	63.96	334.6	298.6	M22	86	88	1.351	26.91	IPE A 360	50.2	14520	811.8	906.8	15.06	29.76	944.3	111.1	171.9	3.84	50.69	26.51	282.0
IPE 360	57.1	360	170	8.0	12.7	18	72.73	334.6	298.6	M22	88	88	1.353	23.70	IPE 360	57.1	16270	903.6	1019	14.95	35.14	1043	122.8	191.1	3.79	54.49	37.32	313.6
IPE O 360	66.0	364	172	9.2	14.7	18	84.13	334.6	298.6	M22	90	90	1.367	20.69	IPE O 360	66.0	19050	1047	1186	15.05	40.21	1251	145.5	226.9	3.86	59.69	55.76	380.3
IPE A 400	57.4	397	180	7.0	12.0	21	73.10	373.0	331.0	M22	94	98	1.464	25.51	IPE A 400	57.4	20290	1022	1144	16.66	35.78	1171	130.1	202.1	4.00	55.60	34.79	432.2
IPE 400	66.3	400	180	8.6	13.5	21	84.46	373.0	331.0	M22	96	98	1.467	22.12	IPE 400	66.3	23130	1156	1307	16.55	42.69	1318	146.4	229.0	3.95	60.20	51.08	490.0
IPE O 400	75.7	404	182	9.7	15.5	21	96.39	373.0	331.0	M22	96	100	1.481	19.57	IPE O 400	75.7	26750	1324	1502	16.66	47.98	1564	171.9	269.1	4.03	65.30	73.10	587.6
IPE A 450	67.2	447	190	7.6	13.1	21	85.55	420.8	378.8	M24	100	102	1.603	23.87	IPE A 450	67.2	29760	1331	1494	18.65	42.26	1502	158.1	245.7	4.19	58.40	45.67	704.9
IPE 450	77.6	450	190	9.4	14.6	21	98.82	420.8	378.8	M24	100	102	1.605	20.69	IPE 450	77.6	33740	1500	1702	18.48	50.85	1676	176.4	276.4	4.12	63.20	66.87	791.0
IPE O 450	92.4	456	192	11.0	17.6	21	117.7	420.8	378.8	M24	102	104	1.622	17.56	IPE O 450	92.4	40920	1795	2046	18.65	59.40	2085.0	217.2	341.0	4.21	70.80	109	997.6
IPE A 500	79.4	497	200	8.4	14.5	21	101.1	468.0	426.0	M24	100	112	1.741	21.94	IPE A 500	79.4	42930	1728	1946	20.61	50.41	1939	193.9	301.6	4.38	62.00	62.78	1125
IPE 500	90.7	500	200	10.2	16.0	21	115.5	468.0	426.0	M24	102	112	1.744	19.23	IPE 500	90.7	48200	1928	2194	20.43	59.87	2142	214.2	335.9	4.31	66.80	89.29	1249
IPE O 500	107.0	506	202	12.0	19.0	21	136.7	468.0	426.0	M24	104	114	1.760	16.40	IPE O 500	107.0	57780	2284	2613	20.56	70.21	2622	259.6	408.5	4.38	74.60	143.5	1548
IPE A 550	92.1	547	210	9.0	15.7	24	117.3	515.6	467.6	M24	106	122	1.875	20.36	IPE A 550	92.1	59980	2193	2475	22.61	60.30	2432	231.6	361.5	4.55	68.52	86.53	1710
IPE 550	106	550	210	11.1	17.2	24	134.4	515.6	467.6	M24	110	122	1.877	17.78	IPE 550	106.0	67120	2441	2787	22.35	72.34	2668	254.1	400.5	4.45	73.62	123.2	1884
IPE O 550	123	556	212	12.7	20.2	24	156.1	515.6	467.6	M24	110	122	1.893	15.45	IPE O 550	123.0	79160	2847	3263	22.52	82.69	3224	304.2	480.5	4.55	81.22	187.5	2302
IPE A 600	108	597	220	9.8	17.5	24	137.0	562.0	514.0	M27	114	118	2.013	18.72	IPE A 600	108.0	82920	2778	3141	24.60	70.14	3116	283.3	442.1	4.77	72.92	118.8	2607
IPE 600	122	600	220	12.0	19.0	24	156.0	562.0	514.0	M27	116	118	2.015	16.45	IPE 600	122.0	92080	3069	3512	24.30	83.78	3387	307.9	485.6	4.66	78.12	165.4	2846
IPE O 600	154	610	224	15.0	24.0	24	196.8	562.0	514.0	M27	118	122	2.045	13.24	IPE O 600	154.0	118300	3879	4471	24.52	104.40	4521	403.6	640.1	4.79	91.12	318.1	3860
IPE 750 x 147	147	753	265	13.2	17.0	17	187.5	719.0	685.0	M27	104	164	2.510	17.06	IPE 750 x 147	147.0	166100	4411	5110	29.76	105.40	5289	399.2	630.8	5.31	67.12	161.5	7141
IPE 750 x 173	173	762	267	14.4	21.6	17	221.3	718.8	684.8	M27	104	166	2.534	14.58	IPE 750 x 173	173.0	205800	5402	6218	30.49	116.40	6873	514.9	809.9	5.57	77.52	273.6	9391
IPE 750 x 196	196	770	268	15.6	25.4	17	250.8	719.2	685.2	M27	106	166	2.552	12.96	IPE 750 x 196	196.0	240300	6241	7174	30.95	127.30	8175	610.1	958.8	5.71	86.32	408.9	11290

Avrupa kesit standart profiller IPN

flange slope: 14%

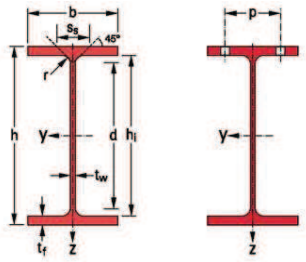


KESİT	ÖLÇÜLER							KESİT ALANI	TASARIM ÖLÇÜLERİ				YÜZEY ALANI		KESİT	STATİK DEĞERLER											
																KUVVETLİ EKSEN y-y					ZAYIF EKSEN z-z						
	G	h	b	t_w	t_f	r_1	r_2	A	d	Ø	p_{min}	p_{max}	A_L	A_G		G	I_y	$W_{el.y}$	$W_{pl.y}$	i_y	A_{vz}	I_z	$W_{el.z}$	$W_{pl.z}$	i_z	s_s	I_t	I_wx10^6
	kg/m	mm	mm	mm	mm	mm	mm	cm^2	mm		mm	mm	m^2/m	m^2/t		kg/m	cm^4	cm^3	cm^3	cm	cm^2	cm^4	cm^3	cm^3	cm	mm	cm^4	cm^6
IPN 120	11.1	120	58	5.1	7.7	5.1	3.1	14.2	92.4	-	-	-	0.439	39.38	IPN 120	11.1	328	54.7	63.6	4.81	6.63	21.5	7.41	12.4	1.23	28.4	2.71	0.69
IPN 140	14.3	140	66	5.7	8.6	5.7	3.4	18.3	109.1	-	-	-	0.502	34.94	IPN 140	14.3	573	81.9	95.4	5.61	8.65	35.2	10.7	17.9	1.40	31.8	4.32	1.54
IPN 160	17.9	160	74	6.3	9.5	6.3	3.8	22.8	125.8	-	-	-	0.575	32.13	IPN 160	17.9	935	117	136	6.4	10.83	54.7	14.8	24.9	1.55	35.2	6.57	3.14
IPN 180	21.9	180	82	6.9	10.4	6.9	4.1	27.9	142.4	-	-	-	0.640	29.22	IPN 180	21.9	1450	161	187	7.2	13.35	81.3	19.8	33.2	1.71	38.6	9.58	5.92
IPN 200	26.2	200	90	7.5	11.3	7.5	4.5	33.4	159.1	-	-	-	0.709	27.04	IPN 200	26.2	2140	214	250	8.0	16.03	117	26.0	43.5	1.87	42.0	13.5	10.5
IPN 220	31.1	220	98	8.1	12.2	8.1	4.9	39.5	175.8	M10	50	56	0.775	24.99	IPN 220	31.1	3060	278	324	8.8	19.06	162	33.1	55.7	2.02	45.4	18.6	17.8
IPN 240	36.2	240	106	8.7	13.1	8.7	5.2	46.1	192.5	M10	54	60	0.844	23.32	IPN 240	36.2	4250	354	412	9.59	22.33	221	41.7	70.0	2.20	48.9	25.0	28.7
IPN 260	41.9	260	113	9.4	14.1	9.4	5.6	53.3	208.9	M12	62	62	0.906	21.65	IPN 260	41.9	5740	442	514	10.4	26.08	288	51.0	85.9	2.32	52.6	33.5	44.1
IPN 280	47.9	280	119	10.1	15.2	10.1	6.1	61.0	225.1	M12	68	68	0.966	20.17	IPN 280	47.9	7590	542	632	11.1	30.18	364	61.2	103	2.45	56.4	44.2	64.6
IPN 300	54.2	300	125	10.8	16.2	10.8	6.5	69.0	241.6	M12	70	74	1.03	19.02	IPN 300	54.2	9800	653	762	11.9	34.58	451	72.2	121	2.56	60.1	56.8	91.8
IPN 320	61.0	320	131	11.5	17.3	11.5	6.9	77.7	257.9	M12	70	80	1.09	17.87	IPN 320	61.0	12510	782	914	12.7	39.26	555	84.7	143	2.67	63.9	72.5	129
IPN 340	68.0	340	137	12.2	18.3	12.2	7.3	86.7	274.3	M12	78	86	1.15	16.90	IPN 340	68.0	15700	923	1080	13.5	44.27	674	98.4	166	2.80	67.6	90.4	176
IPN 360	76.1	360	143	13	19.5	13	7.8	97.0	290.2	M12	78	92	1.21	15.89	IPN 360	76.1	19610	1090	1276	14.2	49.95	818	114.0	194	2.90	71.8	115	240
IPN 380	84.0	380	149	13.7	20.5	13.7	8.2	107	306.7	M16	84	86	1.27	15.12	IPN 380	84.0	24010	1260	1482	15.0	55.55	975	131.0	221	3.02	75.4	141	319
IPN 400	92.4	400	155	14.4	21.6	14.4	8.6	118	322.9	M16	86	92	1.33	14.36	IPN 400	92.4	29210	1460	1714	15.7	61.69	1160	149	253	3.13	79.3	170	420
IPN 450	115	450	170	16.2	24.3	16.2	9.7	147	363.6	M16	92	106	1.48	12.83	IPN 450	115	45850	2040	2400	17.7	77.79	1730	203	345	3.43	88.9	267	791
IPN 500	141	500	185	18	27	18	10.8	179	404.3	M20	102	110	1.63	11.60	IPN 500	141	68740	2750	3240	19.6	95.6	2480	268	456	3.72	98.5	402	1400
IPN 550	166	550	200	19	30	19	11.9	212	445.6	M22	112	118	1.80	10.80	IPN 550	166	99180	3610	4240	21.6	111.30	3490	349	592	4.02	107.3	544	2390

Geniş başlıklı Avrupa kesit profiller

HE A, HE B et HE M 100 - 1000
Euronorm 53-62
HE AA 100 - 1000
HL 1000 / 1100

HEA
HEB
HEM



1	2	3	4	5	6	7	8					14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
KESİT	ÖLÇÜLER						KESİT ALANI	TASARIM ÖLÇÜLERİ					YÜZEY ALANI		KESİT	STATİK DEĞERLER										
																KUVVETLİ EKSEN y-y					ZAYIF EKSEN z-z					
	G	h	b	t_w	t_f	r	A	h_i	d	Ø	pmin	pmax	A_L	A_G		G	I_y	$W_{el.y}$	$W_{pl.y}$	i_y	A_{vz}	I_z	$W_{el.z}$	$W_{pl.z}$	i_z	s_s	I_t	$I_w x10^{-3}$
	kg/m	mm	mm	mm	mm	mm	cm^2	mm	mm		mm	mm	m^2/m	m^2/t		kg/m	cm^4	cm^3	cm^3	cm	cm^2	cm^4	cm^3	cm^3	cm	mm	cm^4	cm^6
HE 100 AA	12.2	91	100	4.2	5.5	12	15.60	80	56	M10	54	58	0.553	45.170	HE 100 AA	12.2	236.5	51.98	58.36	3.89	6.15	92.06	18.41	28.44	2.43	29.26	2.51	1.68
HE 100 A	16.7	96	100	5.0	8.0	12	21.24	80	56	M10	54	58	0.561	33.680	HE 100 A	16.7	349.2	72.76	83.01	4.06	7.56	133.8	26.76	41.14	2.51	35.06	5.24	2.58
HE 100 B	20.4	100	100	6.0	10.0	12	26.04	80	56	M10	56	58	0.567	27.760	HE 100 B	20.4	449.5	89.91	104.20	4.16	9.04	167.3	33.45	51.42	2.53	40.06	9.25	3.38
HE 100 M	41.8	120	106	12.0	20.0	12	53.24	80	56	M10	62	64	0.619	14.820	HE 100 M	41.8	1143.0	190.4	235.80	4.63	18.04	399.2	75.31	116.30	2.74	66.06	68.21	9.93
HE 120 AA	14.6	109	120	4.2	5.5	12	18.55	98	74	M12	58	68	0.669	45.940	HE 120 AA	14.6	413.4	75.85	84.12	4.72	6.90	158.8	26.47	40.62	2.93	29.26	2.78	4.24
HE 120 A	19.9	114	120	5.0	8.0	12	25.34	98	74	M12	58	68	0.677	34.060	HE 120 A	19.9	606.2	106.3	119.5	4.89	8.46	230.9	38.48	58.85	3.02	35.06	5.99	6.47
HE 120 B	26.7	120	120	6.5	11.0	12	34.01	98	74	M12	60	68	0.686	25.710	HE 120 B	26.7	864.4	144.1	165.2	5.04	10.96	317.5	52.92	80.97	3.06	42.56	13.84	9.41
HE 120 M	52.1	140	126	12.5	21.0	12	66.41	98	74	M12	66	74	0.738	14.160	HE 120 M	52.1	2018.0	288.2	350.6	5.51	21.15	702.8	111.60	171.60	3.25	68.56	91.66	24.79
HE 140 AA	18.1	128	140	4.3	6.0	12	23.02	116	92	M16	64	76	0.787	43.530	HE 140 AA	18.1	719.5	112.4	123.8	5.59	7.92	274.8	39.26	59.93	3.45	30.36	3.54	10.21
HE 140 A	24.7	133	140	5.5	8.5	12	31.42	116	92	M16	64	76	0.794	32.210	HE 140 A	24.7	1033.0	155.4	173.5	5.73	10.12	389.3	55.62	84.85	3.52	36.56	8.13	15.06
HE 140 B	33.7	140	140	7.0	12.0	12	42.96	116	92	M16	66	76	0.805	23.880	HE 140 B	33.7	1509.0	215.6	245.4	5.93	13.08	549.7	78.52	119.80	3.58	45.06	20.06	22.48
HE 140 M	63.2	160	146	13.0	22.0	12	80.56	116	92	M16	72	82	0.857	13.560	HE 140 M	63.2	3291.0	411.4	493.8	6.39	24.46	1144.0	156.80	240.50	3.77	71.06	120.00	54.33
HE 160 AA	23.8	148	160	4.5	7.0	15	30.36	134	104	M20	76	84	0.901	37.810	HE 160 AA	23.8	1283.0	173.4	190.4	6.50	10.38	478.7	59.84	91.36	3.97	36.07	6.33	23.75
HE 160 A	30.4	152	160	6.0	9.0	15	38.77	134	104	M20	78	84	0.906	29.780	HE 160 A	30.4	1673.0	220.1	245.1	6.57	13.21	615.6	76.95	117.6	3.98	41.57	12.19	31.41
HE 160 B	42.6	160	160	8.0	13.0	15	54.25	134	104	M20	80	84	0.918	21.560	HE 160 B	42.6	2492.0	311.5	354.0	6.78	17.59	889.2	111.2	170.0	4.05	51.57	31.24	47.94
HE 160 M	76.2	180	166	14.0	23.0	15	97.05	134	104	M20	86	90	0.970	12.740	HE 160 M	76.2	5098.0	566.5	674.6	7.25	30.81	1759.0	211.9	325.5	4.26	77.57	162.40	108.10
HE 180 AA	28.7	167	180	5.0	7.5	15	36.53	152	122	M24	84	92	1.018	35.510	HE 180 AA	28.7	1967.0	235.6	258.2	7.34	12.16	730.0	81.1	123.6	4.47	37.57	8.33	46.36
HE 180 A	35.5	171	180	6.0	9.5	15	45.25	152	122	M24	86	92	1.024	28.830	HE 180 A	35.5	2510.0	293.6	324.9	7.45	14.47	924.6	102.7	156.5	4.52	42.57	14.80	60.21
HE 180 B	51.2	180	180	8.5	14.0	15	65.25	152	122	M24	88	92	1.037	20.250	HE 180 B	51.2	3831.0	425.7	481.4	7.66	20.24	1363.0	151.4	231.0	4.57	54.07	42.16	93.75
HE 180 M	88.9	200	186	14.5	24.0	15	113.30	152	122	M24	94	98	1.089	12.250	HE 180 M	88.9	7483.0	748.3	883.4	8.13	34.65	2580	277.4	425.2	4.77	80.07	203.30	199.30
HE 200 AA	34.6	186	200	5.5	8.0	18	44.13	170	134	M27	96	100	1.130	32.620	HE 200 AA	34.6	2944.0	316.6	347.1	8.17	15.45	1068.0	106.8	163.2	4.92	42.59	12.69	84.49
HE 200 A	42.3	190	200	6.5	10.0	18	53.83	170	134	M27	98	100	1.136	26.890	HE 200 A	42.3	3692.0	388.6	429.5	8.28	18.08	1336.0	133.6	203.8	4.98	47.59	20.98	108.0
HE 200 B	61.3	200	200	9.0	15.0	18	78.08	170	134	M27	100	100	1.151	18.780	HE 200 B	61.3	5696.0	569.6	642.5	8.54	24.83	2003.0	200.3	305.8	5.07	60.09	59.28	171.1
HE 200 M	103.0	220	206	15.0	25.0	18	131.30	170	134	M27	106	106	1.203	11.670	HE 200 M	103.0	10640.0	967.4	1135.0	9.00	41.03	3651.0	354.5	543.2	5.27	86.09	259.40	346.3

HE 220 AA	40.4	205	220	6.0	8.5	18	51.46	188	152	M27	98	118	1.247	30.870	HE 220 AA	40.4	4170.0	406.9	445.5	9.00	17.63	1510.0	137.3	209.3	5.42	44.09	15.93	145.6
HE 220 A	50.5	210	220	7.0	11.0	18	64.34	188	152	M27	98	118	1.255	24.850	HE 220 A	50.5	5410.0	515.2	568.5	9.17	20.67	1955	177.7	270.6	5.51	50.09	28.46	193.3
HE 220 B	71.5	220	220	9.5	16.0	18	91.04	188	152	M27	100	118	1.270	17.770	HE 220 B	71.5	8091.0	735.5	827.0	9.43	27.92	2843.0	258.5	393.9	5.59	62.59	76.57	295.4
HE 220 M	117.0	240	226	15.5	26.0	18	149.40	188	152	M27	108	124	1.322	11.270	HE 220 M	117.0	14600.0	1217.0	1419.0	9.89	45.31	5012.0	443.5	678.6	5.79	88.59	315.30	572.7
HE 240 AA	47.4	224	240	6.5	9.0	21	60.38	206	164	M27	104	138	1.359	28.670	HE 240 AA	47.4	5835.0	521.0	570.6	9.83	21.54	2077	173.1	264.4	5.87	49.10	22.98	239.6
HE 240 A	60.3	230	240	7.5	12.0	21	76.84	206	164	M27	104	138	1.369	22.700	HE 240 A	60.3	7763.0	675.1	744.6	10.05	25.18	2769	230.7	351.7	6.00	56.10	41.55	328.5
HE 240 B	83.2	240	240	10.0	17.0	21	106.00	206	164	M27	108	138	1.384	16.630	HE 240 B	83.2	11260.0	938.3	1053.0	10.31	33.23	3923	326.9	498.4	6.08	68.60	102.70	486.9
HE 240 M	157.0	270	248	18.0	32.0	21	199.60	206	164	M27	116	146	1.460	9.318	HE 240 M	157.0	24290.0	1799.0	2117.0	11.03	60.07	8153	657.5	1006.0	6.39	106.60	627.90	1152.0
HE 260 AA	54.1	244	260	6.5	9.5	24	68.97	225	177	M27	110	158	1.474	27.220	HE 260 AA	54.1	7981.0	654.1	714.5	10.76	24.75	2788	214.5	327.7	6.36	53.62	30.31	382.6
HE 260 A	68.2	250	260	7.5	12.5	24	86.82	225	177	M27	110	158	1.484	21.770	HE 260 A	68.2	10450.0	836.4	919.8	10.97	28.76	3668	282.1	430.2	6.50	60.62	52.37	516.4
HE 260 B	93.0	260	260	10.0	17.5	24	118.40	225	177	M27	114	158	1.499	16.120	HE 260 B	93.0	14920.0	1148.0	1283.0	11.22	37.59	5135	395.0	602.2	6.58	73.12	123.80	753.7
HE 260 M	172	290	268	18.0	32.5	24	219.60	225	177	M27	122	166	1.575	9.133	HE 260 M	172.0	31310.0	2159.0	2524.0	11.94	66.89	10450	779.7	1192.0	6.90	111.10	719.00	1728.0
HE 280 AA	61.2	264	280	7.0	10.0	24	78.02	244	196	M27	110	178	1.593	26.010	HE 280 AA	61.2	10560.0	799.8	873.1	11.63	27.52	3664	261.7	399.4	6.85	55.12	36.22	590.1
HE 280 A	76.4	270	280	8.0	13.0	24	97.26	244	196	M27	112	178	1.603	20.990	HE 280 A	76.4	13670.0	1013.0	1112	11.86	31.74	4763	340.2	518.1	7.00	62.12	62.10	785.4
HE 280 B	103	280	280	10.5	18.0	24	131.40	244	196	M27	114	178	1.618	15.690	HE 280 B	103.0	19270.0	1376.0	1534	12.11	41.09	6595	471.0	717.6	7.09	74.62	143.70	1130.0
HE 280 M	189.0	310	288	18.5	33.0	24	240.20	244	196	M27	122	186	1.694	8.984	HE 280 M	189.0	39550.0	2551.0	2966.0	12.83	72.03	13160	914.1	1397.0	7.40	112.60	807.30	2520.0
HE 300 AA	69.8	283	300	7.5	10.5	27	88.91	262	208	M27	116	198	1.705	24.420	HE 300 AA	69.8	13800.0	976	1065	12.46	32.37	4734	315.6	482.3	7.30	60.13	49.35	877.2
HE 300 A	88	290	300	8.5	14.0	27	112.50	262	208	M27	118	198	1.717	19.430	HE 300 A	88.3	18260.0	1260	1383	12.74	37.28	6310	420.6	641.2	7.49	68.13	85.17	1200
HE 300 B	117.0	300	300	11.0	19.0	27	149.10	262	208	M27	120	198	1.732	14.800	HE 300 B	117.0	25170.0	1678.0	1869.0	12.99	47.43	8563	570.9	870.1	7.58	80.63	185.00	1688
HE 300 M	238.0	340	310	21.0	39.0	27	303.10	262	208	M27	132	208	1.832	7.699	HE 300 M	238.0	59200.0	3482	4078	13.98	90.53	19400	1252.0	1913.0	8.00	130.60	1408.00	4386
HE 320 AA	74	301	300	8.0	11.0	27	94.58	279	225	M27	118	198	1.740	23.430	HE 320 AA	74.2	16450.0	1093	1196	13.19	35.40	4959	330.6	505.7	7.24	61.63	55.87	1041
HE 320 A	97.6	310	300	9.0	15.5	27	124.40	279	225	M27	118	198	1.756	17.980	HE 320 A	97.6	22930.0	1479.0	1628	13.58	41.13	6985	465.7	709.7	7.49	71.63	108.00	1512
HE 320 B	127.0	320	300	11.5	20.5	27	161.30	279	225	M27	122	198	1.771	13.980	HE 320 B	127.0	30820.0	1926	2149	13.82	51.77	9239	615.9	939.1	7.57	84.13	225.10	2069
HE 320 M	245	359	309	21.0	40.0	27	312.00	279	225	M27	132	204	1.866	7.616	HE 320 M	245.0	68130.0	3796	4435	14.78	94.85	19710	1276.0	1951.0	7.95	132.60	1501.00	5004
HE 340 AA	78.9	320	300	8.5	11.5	27	100.50	297	243	M27	118	198	1.777	22.520	HE 340 AA	78.9	19550.0	1222	1341	13.95	38.69	5185	345.6	529.3	7.18	63.13	63.07	1231
HE 340 A	105.0	330	300	9.5	16.5	27	133.50	297	243	M27	118	198	1.795	17.130	HE 340 A	105.0	27690.0	1678	1850	14.40	44.95	7436	495.7	755.9	7.46	74.13	127.20	1824
HE 340 B	134	340	300	12.0	21.5	27	170.90	297	243	M27	122	198	1.810	13.490	HE 340 B	134.0	36660.0	2156	2408	14.65	56.09	9690	646.0	985.7	7.53	86.63	257.20	2454
HE 340 M	248	377	309	21.0	40.0	27	315.80	297	243	M27	132	204	1.902	7.670	HE 340 M	248.0	76370.0	4052	4718	15.55	98.63	19710	1276.0	1953.0	7.90	132.60	1506.00	5584
HE 360 AA	84	339	300	9.0	12.0	27	106.60	315	261	M27	118	198	1.814	21.670	HE 360 AA	83.7	23040.0	1359	1495	14.70	42.17	5410	360.7	553.0	7.12	64.63	70.99	1444
HE 360 A	112	350	300	10.0	17.5	27	142.80	315	261	M27	120	198	1.834	16.360	HE 360 A	112.0	33090.0	1891	2088	15.22	48.96	7887	525.8	802.3	7.43	76.63	148.80	2177
HE 360 B	142	360	300	12.5	22.5	27	180.60	315	261	M27	122	198	1.849	13.040	HE 360 B	142.0	43190.0	2400	2683	15.46	60.60	10140	676.1	1032.0	7.49	89.13	292.50	2883
HE 360 M	250.0	395	308	21.0	40.0	27	318.80	315	261	M27	132	204	1.934	7.730	HE 360 M	250.0	84870.0	4297	4989	16.32	102.40	19520	1268.0	1942.0	7.83	132.60	1507.00	6137
HE 400 AA	92	378	300	9.5	13.0	27	117.70	352	298	M27	118	198	1.891	20.460	HE 400 AA	92.4	31250.0	1654	1824	16.30	47.95	5861	390.8	599.7	7.06	67.13	84.69	1948
HE 400 A	125	390	300	11.0	19.0	27	159.00	352	298	M27	120	198	1.912	15.320	HE 400 A	125.0	45070.0	2311	2562	16.84	57.33	8564	570.9	872.9	7.34	80.63	189.00	2942
HE 400 B	155.0	400	300	13.5	24.0	27	197.80	352	298	M27	124	198	1.927	12.410	HE 400 B	155.0	57680.0	2884	3232	17.08	69.98	10820	721.3	1104.0	7.40	93.13	355.70	3817
HE 400 M	256	432	307	21.0	40.0	27	325.80	352	298	M27	132	202	2.004	7.835	HE 400 M	256.0	104100.0	4820	5571	17.88	110.20	19340	1260.0	1934.0	7.70	132.60	1515.00	7410
HE 450 AA	100	425	300	10.0	13.5	27	127.10	398	344	M27	120	198	1.984	19.890	HE 450 AA	99.7	41890.0	1971	2183	18.16	54.70	6088	405.8	624.4	6.92	68.63	95.61	2572

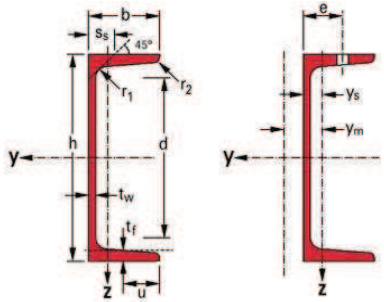
HE 450 A	140.0	440	300	11.5	21.0	27	178.00	398	344	M27	122	198	2.011	14.390	HE 450 A	140.0	63720.0	2896	3216	18.92	65.78	9465	631.0	965.5	7.29	85.13	243.80	4148
HE 450 B	171	450	300	14.0	26.0	27	218.00	398	344	M27	124	198	2.026	11.840	HE 450 B	171.0	79890.0	3551	3982	19.14	79.66	11720	781.4	1198.0	7.33	97.63	440.50	5258
HE 450 M	263	478	307	21.0	40.0	27	335.40	398	344	M27	132	202	2.096	7.959	HE 450 M	263.0	131500.0	5501	6331	19.80	119.80	19340	1260.0	1939.0	7.59	132.60	1529.00	9251
HE 500 AA	107	472	300	10.5	14.0	27	136.90	444	390	M27	120	198	2.077	19.330	HE 500 AA	107.0	54640.0	2315	2576	19.98	61.91	6314	420.9	649.3	6.79	70.13	107.70	3304
HE 500 A	155	490	300	12.0	23.0	27	197.50	444	390	M27	122	198	2.110	13.600	HE 500 A	155.0	86970.0	3550	3949	20.98	74.72	10370	691.1	1059.0	7.24	89.63	309.30	5643
HE 500 B	187	500	300	14.5	28.0	27	238.60	444	390	M27	124	198	2.125	11.340	HE 500 B	187.0	107200.0	4287	4815	21.19	89.82	12620	841.6	1292.0	7.27	102.10	538.40	7018
HE 500 M	270	524	306	21.0	40.0	27	344.30	444	390	M27	132	202	2.184	8.079	HE 500 M	270.0	161900.0	6180	7094	21.69	129.50	19150	1252.0	1932.0	7.46	132.60	1539.00	11190
HE 550 AA	120	522	300	11.5	15.0	27	152.80	492	438	M27	122	198	2.175	18.130	HE 550 AA	120.0	72870.0	2792	3128	21.84	72.66	6767	451.1	698.6	6.65	73.13	133.70	4338
HE 550 A	166	540	300	12.5	24.0	27	211.80	492	438	M27	122	198	2.209	13.290	HE 550 A	166.0	111900.0	4146	4622	22.99	83.72	10820	721.3	1107.0	7.15	92.13	351.50	7189
HE 550 B	199	550	300	15.0	29.0	27	254.10	492	438	M27	124	198	2.224	11.150	HE 550 B	199.0	136700.0	4971	5591	23.20	100.10	13080	871.8	1341.0	7.17	104.60	600.30	8856
HE 550 M	278	572	306	21.0	40.0	27	354.40	492	438	M27	132	202	2.280	8.195	HE 550 M	278.0	198000.0	6923	7933	23.64	139.60	19160	1252.0	1937.0	7.35	132.60	1554.00	13520
HE 600 AA	129	571	300	12.0	15.5	27	164.10	540	486	M27	122	198	2.272	17.640	HE 600 AA	129.0	91900.0	3218	3623	23.66	81.29	6993	466.2	724.5	6.53	74.63	149.80	5381
HE 600 A	178	590	300	13.0	25.0	27	226.50	540	486	M27	122	198	2.308	12.980	HE 600 A	178.0	141200.0	4787	5350	24.97	93.21	11270	751.4	1156.0	7.05	94.63	397.80	8978
HE 600 B	212	600	300	15.5	30.0	27	270.00	540	486	M27	126	198	2.323	10.960	HE 600 B	212.0	171000.0	5701	6425	25.17	110.80	13530	902.0	1391.0	7.08	107.10	667.20	10970
HE 600 M	285	620	305	21.0	40.0	27	363.70	540	486	M27	132	200	2.372	8.308	HE 600 M	285.0	237400.0	7660	8772	25.55	149.70	18980	1244.0	1930.0	7.22	132.60	1564.00	15910
HE 600 x 337	337	632	310	25.5	46.0	27	429.20	540	486	M27	138	202	2.407	7.144	HE 600 x 337	337.0	283200.0	8961	10380	25.69	180.50	22940	1480.0	2310.0	7.31	149.10	2451.00	19610
HE 600 x 399	399	648	315	30.0	54.0	27	508.50	540	486	M27	142	208	2.450	6.137	HE 600 x 399	399.0	344600.0	10640	12460	26.03	213.60	28280	1796.0	2814.0	7.46	169.60	3966.00	24810
HE 650 AA	138	620	300	12.5	16.0	27	175.80	588	534	M27	122	198	2.369	17.170	HE 650 AA	138.0	113900.0	3676	4160	25.46	90.40	7221	481.4	750.7	6.41	76.13	167.50	6567
HE 650 A	190	640	300	13.5	26.0	27	241.60	588	534	M27	124	198	2.407	12.690	HE 650 A	190.0	175200.0	5474	6136	26.93	103.20	11720	781.6	1205.0	6.97	97.13	448.30	11030
HE 650 B	225	650	300	16.0	31.0	27	286.30	588	534	M27	126	198	2.422	10.770	HE 650 B	225.0	210600.0	6480	7320	27.12	122.00	13980	932.3	1441.0	6.99	109.60	739.20	13360
HE 650 M	293	668	305	21.0	40.0	27	373.70	588	534	M27	132	200	2.468	8.411	HE 650 M	293.0	281700.0	8433	9657	27.45	159.70	18980	1245.0	1936.0	7.13	132.60	1579.00	18650
HE 650 x 343	343	680	309	25.0	46.0	27	437.50	588	534	M27	138	202	2.500	7.278	HE 650 x 343	343.0	333700.0	9815	11350	27.62	189.60	22720	1470.0	2300.0	7.21	148.60	2442.00	22730
HE 650 x 407	407	696	314	29.5	54.0	27	518.80	588	534	M27	142	206	2.543	6.243	HE 650 x 407	407.0	405400.0	11650	13620	27.95	224.80	28020	1785.0	2803.0	7.35	169.10	3958.00	28710
HE 700 AA	150	670	300	13.0	17.0	27	190.90	636	582	M27	122	198	2.468	16.460	HE 700 AA	150.0	142700.0	4260	4840	27.34	100.30	7673	511.5	799.7	6.34	78.63	195.20	8155
HE 700 A	204	690	300	14.5	27.0	27	260.50	636	582	M27	124	198	2.505	12.250	HE 700 A	204.0	215300.0	6241	7032	28.75	117.00	12180	811.9	1257.0	6.84	100.10	513.90	13350
HE 700 B	241	700	300	17.0	32.0	27	306.40	636	582	M27	126	198	2.520	10.480	HE 700 B	241.0	256900.0	7340	8327	28.96	137.10	14440	962.7	1495.0	6.87	112.60	830.90	16060
HE 700 M	301	716	304	21.0	40.0	27	383.00	636	582	M27	132	200	2.560	8.513	HE 700 M	301.0	329300.0	9198	10540	29.32	169.80	18800	1237.0	1929.0	7.01	132.60	1589.00	21400
HE 700 x 352	352	728	308	25.0	46.0	27	448.60	636	582	M27	138	200	2.592	7.359	HE 700 x 352	352.0	389700.0	10710	12390	29.47	201.60	22510	1461.0	2293.0	7.08	148.60	2461.00	26050
HE 700 x 418	418	744	313	29.5	54.0	27	531.90	636	582	M27	142	206	2.635	6.310	HE 700 x 418	418.0	472500.0	12700	14840	29.80	239.00	27760	1774.0	2797.0	7.22	169.10	3989.00	32850
HE 800 AA	172	770	300	14.0	18.0	30	218.50	734	674	M27	130	198	2.660	15.510	HE 800 AA	172.0	208900.0	5426	6225	30.92	123.80	8134	542.2	856.6	6.10	85.15	256.80	11450
HE 800 A	224	790	300	15.0	28.0	30	285.80	734	674	M27	130	198	2.698	12.030	HE 800 A	224.0	303400.0	7682	8699	32.58	138.80	12640	842.6	1312.0	6.65	106.10	596.90	18290
HE 800 B	262	800	300	17.5	33.0	30	334.20	734	674	M27	134	198	2.713	10.340	HE 800 B	262.0	359100.0	8977	10230	32.78	161.80	14900	993.6	1553.0	6.68	118.60	946.00	21840
HE 800 M	317	814	303	21.0	40.0	30	404.30	734	674	M27	138	198	2.746	8.655	HE 800 M	317.0	442600.0	10870	12490	33.09	194.30	18630	1230.0	1930.0	6.79	136.10	1646.00	27780
HE 800 x 373	373	826	308	25.0	46.0	30	474.60	734	674	M27	144	200	2.782	7.469	HE 800 x 373	373.0	523900.0	12690	14700	33.23	230.30	22530	1463.0	2311.0	6.89	152.10	2554.00	34070
HE 800 x 444	444	842	313	30.0	54.0	30	566.00	734	674	M27	148	206	2.824	6.357	HE 800 x 444	444.0	634500.0	15070	17640	33.48	276.50	27800	1776.0	2827.0	7.01	173.10	4180.00	42840
HE 900 AA	198	870	300	15.0	20.0	30	252.20	830	770	M27	130	198	2.858	14.440	HE 900 AA	198.0	301100.0	6923	7999	34.55	147.20	9041	602.8	957.7	5.99	90.15	334.90	16260
HE 900 A	252	890	300	16.0	30.0	30	320.50	830	770	M27	132	198	2.896	11.510	HE 900 A	252.0	422100.0	9485	10810	36.29	163.30	13550	903.2	1414.0	6.50	111.10	736.80	24960
HE 900 B	291	900	300	18.5	35.0	30	371.30	830	770	M27	134	198	2.911	9.990	HE 900 B	291.0	494100.0	10980	12580	36.48	188.80	15820	1054.0	1658.0	6.53	123.60	1137.00	29460
HE 900 M	333	910	302	21.0	40.0	30	423.60	830	770	M27	138	198	2.934	8.824	HE 900 M	333.0	570400.0	12540	14440	36.70	214.40	18450	1222.0	1929.0	6.60	136.10	1671.00	34750

HE 900 x 391	391	922	307	25.0	46.0	30	497.70	830	770	M27	144	200	2.970	7.604	HE 900 x 391	391.0	674300.0	14630	16990	36.81	254.30	22320	1454.0	2312.0	6.70	152.10	2597.00	42560
HE 900 x 466	466	938	312	30.0	54.0	30	593.70	830	770	M27	148	204	3.012	6.464	HE 900 x 466	466.0	814900.0	17380	20380	37.05	305.30	27560	1767.0	2832.0	6.81	173.10	4256.00	53400
HE 1000 AA	222	970	300	16.0	21.0	30	282.20	928	868	M27	132	198	3.056	13.800	HE 1000 AA	222.0	406500.0	8380	9777	37.95	172.20	9501	633.4	1016.0	5.80	93.15	403.40	21280
HE 1000 A	272	990	300	16.5	31.0	30	346.80	928	868	M27	132	198	3.095	11.370	HE 1000 A	272.0	553800.0	11190	12820	39.96	184.60	14000	933.6	1470.0	6.35	113.60	822.40	32070
HE 1000 B	314	1000	300	19.0	36.0	30	400.00	928	868	M27	134	198	3.110	9.905	HE 1000 B	314.0	644700.0	12890	14860	40.15	212.50	16280	1085.0	1716.0	6.38	126.10	1254.00	37640
HE 1000 M	349	1008	302	21.0	40.0	30	444.20	928	868	M27	138	198	3.130	8.978	HE 1000 M	349.0	722300.0	14330	16570	40.32	235.00	18460	1222.0	1940.0	6.45	136.10	1701.00	43020
HE 1000 x 393	393	1016	303	24.4	43.9	30	500.20	928	868	M27	144	196	3.144	8.006	HE 1000 x 393	393.0	807700.0	15900	18540	40.18	271.30	20500	1353.0	2168.0	6.40	147.30	2332.00	48080
HE 1000 x 409	409	1020	306	25.0	46.0	30	521.20	928	868	M27	144	198	3.162	7.729	HE 1000 x 409	409.0	850800.0	16680	19450	40.40	278.80	22120	1446.0	2313.0	6.51	152.10	2642.00	52100
HE 1000 x 488	488	1036	311	30.0	54.0	30	622.00	928	868	M27	148	204	3.204	6.563	HE 1000 x 488	488.0	########	19820	23300	40.62	334.70	27320	1757.0	2837.0	6.63	173.10	4334.00	65270
HE 1000 x 579	579	1056	316	35.0	64.0	30	737.00	928	868	M27	154	208	3.254	5.625	HE 1000 x 579	579.0	########	23590	27950	41.11	393.30	34040	2154.0	3498.0	6.80	198.10	7102.00	82800
HL 920 x 342	342	912	418	19.3	32.0	24	436.10	848	800	M27	126	312	3.420	9.980	HL 920 x 342	342.0	624900.0	13700	15450	37.85	190.10	39010	1867.0	2882.0	9.46	111.40	1193.00	75410
HL 920 x 365	365	916	419	20.3	34.3	24	464.40	847.4	799.4	M27	128	314	3.430	9.400	HL 920 x 365	365.0	670500.0	14640	16520	38.00	200.40	42120	2011.0	3106.0	9.52	117.00	1446.00	81730
HL 920 x 387	387	921	420	21.3	36.6	24	493.00	847.8	799.8	M27	128	314	3.440	8.880	HL 920 x 387	387.0	718300.0	15600	17630	38.17	210.90	45280	2156.0	3332.0	9.58	122.60	1734.00	88370
HL 920 x 417	417	928	422	22.5	39.9	24	532.50	848.2	800.2	M27	130	316	3.460	8.270	HL 920 x 417	417.0	787600.0	16970	19210	38.46	223.90	50070	2373.0	3668.0	9.70	130.40	2200.00	98540
HL 920 x 446	446	933	423	24.0	42.7	24	569.60	847.6	799.6	M27	130	318	3.470	7.760	HL 920 x 446	446.0	846800.0	18150	20600	38.56	239.10	53980	2552.0	3951.0	9.73	137.50	2685.00	106740
HL 920 x 488	488	942	422	25.9	47.0	24	621.30	848	800	M27	132	316	3.480	7.130	HL 920 x 488	488.0	935390.0	19860	22615	38.80	259.30	59010	2797.0	4336.0	9.75	148.00	3514.00	117890
HL 920 x 534	534	950	425	28.4	51.1	24	680.10	847.8	799.8	M27	136	320	3.500	6.560	HL 920 x 534	534.0	########	21710	24830	38.94	284.80	65560	3085.0	4796.0	9.82	158.70	4542.00	132070
HL 920 x 585	585	960	427	31.0	55.9	24	745.30	848.2	800.2	M27	138	322	3.520	6.020	HL 920 x 585	585.0	########	23814	27363	39.16	312.00	72770	3408.0	5310.0	9.88	170.90	5932.00	148220
HL 920 x 653	653	972	431	34.5	62.0	24	831.90	848	800	M27	144	320	3.560	5.450	HL 920 x 653	653.0	########	26590	30730	39.41	348.70	83050	3854.0	6022.0	9.99	186.60	8124.00	171280
HL 920 x 784	784	996	437	40.9	73.9	24	997.70	848.2	800.2	M27	152	326	3.620	4.620	HL 920 x 784	784.0	########	31980	37340	39.95	417.60	103300	4728.0	7424.0	10.18	216.80	########	218490
HL 920 x 967	967	1028	446	50.0	89.9	24	1231.00	848.2	800.2	M27	160	334	3.700	3.830	HL 920 x 967	967.0	########	39540	46810	40.64	517.10	133900	6003.0	9486.0	10.43	257.90	########	292450
HL 1000 x 296	296	982	400	16.5	27.0	30	376.80	928	868	M27	134	294	3.479	11.760	HL 1000 x 296	296.0	618700.0	12600	14220	40.52	181.50	28850	1443.0	2235.0	8.75	105.60	756.90	65670
HL 1000 A	321	990	400	16.5	31.0	30	408.80	928	868	M27	134	294	3.495	10.890	HL 1000 A	321.0	696400.0	14070	15800	41.27	184.60	33120	1656.0	2555.0	9.00	113.60	1021.00	76030
HL 1000 B	371	1000	400	19.0	36.0	30	472.00	928	868	M27	136	294	3.510	9.474	HL 1000 B	371.0	812100.0	16240	18330	41.48	212.50	38480	1924.0	2976.0	9.03	126.10	1565.00	89210
HL 1000 M	412	1008	402	21.0	40.0	30	524.20	928	868	M27	142	290	3.530	8.580	HL 1000 M	412.0	909800.0	18050	20440	41.66	235.00	43410	2160.0	3348.0	9.10	136.10	2128.00	101460
HL 1000 x 477	477	1018	404	25.5	45.0	30	608.00	928	868	M27	146	292	3.549	7.437	HL 1000 x 477	477.0	########	20570	23530	41.50	282.80	49610	2456.0	3838.0	9.03	150.60	3159.00	117050
HL 1000 x 554	554	1032	408	29.5	52.0	30	705.80	928	868	M27	150	296	3.585	6.471	HL 1000 x 554	554.0	########	23880	27500	41.79	328.00	59100	2897.0	4547.0	9.15	168.60	4860.00	141330
HL 1000 x 642	642	1048	412	34.0	60.0	30	817.60	928	868	M27	154	300	3.624	5.647	HL 1000 x 642	642.0	########	27680	32100	42.12	379.60	70280	3412.0	5379.0	9.27	189.10	7440.00	170670
HL 1000 x 748	748	1068	417	39.0	70.0	30	953.40	928	868	M27	160	304	3.674	4.909	HL 1000 x 748	748.0	########	32430	37880	42.62	438.90	85111	4082.0	6459.0	9.45	214.10	########	210650
HL 1000 x 883	883	1092	424	45.5	82.0	30	1125.30	928	868	M27	166	312	3.737	4.231	HL 1000 x 883	883.0	########	38390	45260	43.16	516.50	105000	4952.0	7874.0	9.66	244.60	########	265670
HL 1100 A	343	1090	400	18.0	31.0	20	436.50	1028	988	M27	116	294	3.710	10.830	HL 1100 A	343.0	867400.0	15920	18060	44.58	206.50	33120	1656.0	2568.0	8.71	103.40	1037.00	92710
HL 1100 B	390	1100	400	20.0	36.0	20	497.00	1028	988	M27	118	294	3.726	9.549	HL 1100 B	390.0	########	18280	20780	44.98	230.60	38480	1924.0	2988.0	8.80	115.40	1564.00	108680
HL 1100 M	433	1108	402	22.0	40.0	20	551.20	1028	988	M27	122	290	3.746	8.657	HL 1100 M	433.0	########	20320	23160	45.19	254.40	43410	2160.0	3362.0	8.87	125.40	2130.00	123500
HL 1100 R	499	1118	405	26.0	45.0	20	635.20	1028	988	M27	126	294	3.770	7.560	HL 1100 R	499.0	########	23150	26600	45.14	300.40	49980	2468.0	3870.0	8.87	139.40	3135.00	143410

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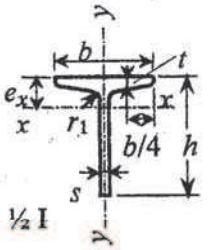
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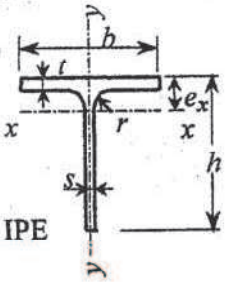
KESİT	ÖLÇÜLER						KESİT ALANI	TASARIM ÖLÇÜLERİ				YÜZEY ALANI		KESİT	STATİK DEĞERLER												
															KUVVETLİ EKSEN y-y					ZAYIF EKSEN z-z								
	G	h	b	t_w	t_f	r_1	r_2	A	d	Ø	e_{min}	e_{max}	A_L	A_G		G	I_y	$W_{el.y}$	$W_{pl.y}$	i_y	A_{vz}	I_z	$W_{el.z}$	$W_{pl.z}$	i_z	s_s	I_t	I_wx10	y_s	y_m
	kg/m	mm	mm	mm	mm	mm	mm	cm^2	mm		mm	mm	m^2/m	m^2/t		kg/m	cm^4	cm^3	cm^3	cm	cm^2	cm^4	cm^3	cm^3	cm	mm	cm^4	cm^6	cm	cm
UPN 100	10.6	100	50	6	8.5	8.5	4.5	13.5	64	-	-	-	0.372	35.1	UPN 100	10.6	206	41.2	49	3.91	6.46	29.3	8.49	16.2	1.47	20.3	2.81	0.41	1.55	2.93
UPN 120	13.4	120	55	7	9	9	4.5	17.0	82	-	-	-	0.434	32.52	UPN 120	13.4	364	60.7	72.6	4.62	8.8	43.2	11.1	21.2	1.59	22.2	4.15	0.9	1.6	3.03
UPN 140	16.0	140	60	7	10	10	5	20.4	98	M12	33	37	0.489	30.54	UPN 140	16.0	605	86.4	103	5.45	10.41	62.7	14.8	28.3	1.75	23.9	5.68	1.8	1.75	3.37
UPN 160	18.8	160	65	7.5	10.5	10.5	5.5	24.0	115	M12	34	42	0.546	28.98	UPN 160	18.8	925	116	138	6.21	12.6	85.3	18.3	35.2	1.89	25.3	7.39	3.26	1.84	3.56
UPN 180	22.0	180	70	8	11	11	5.5	28.0	133	M16	38	41	0.611	27.8	UPN 180	22.0	1350	150	179	6.95	15.09	114	22.4	42.9	2.02	26.7	9.55	5.57	1.92	3.75
UPN 200	25.3	200	75	8.5	11.5	11.5	6	32.2	151	M16	39	46	0.661	26.15	UPN 200	25.3	1910	191	228	7.7	17.71	148	27	51.8	2.14	28.1	11.9	9.07	2.01	3.94
UPN 220	29.4	220	80	9	12.5	12.5	6.5	37.4	167	M16	40	51	0.718	24.46	UPN 220	29.4	2690	245	292	8.48	20.62	197	33.6	64.1	2.3	30.3	16	14.6	2.14	4.2
UPN 240	33.2	240	85	9.5	13	13	6.5	42.3	184	M20	46	50	0.775	23.34	UPN 240	33.2	3600	300	358	9.22	23.71	248	39.6	75.7	2.42	31.7	19.7	22.1	2.23	4.39
UPN 260	37.9	260	90	10	14	14	7	48.3	200	M22	50	52	0.834	22	UPN 260	37.9	4820	371	442	9.99	27.12	317	47.7	91.6	2.56	33.9	25.5	33.3	2.36	4.66
UPN 280	41.8	280	95	10	15	15	7.5	53.3	216	M22	52	57	0.89	21.27	UPN 280	41.8	6280	448	532	10.9	29.28	399	57.2	109	2.74	35.6	31	48.5	2.53	5.02
UPN 300	46.2	300	100	10	16	16	8	58.8	232	M24	55	59	0.95	20.58	UPN 300	46.2	8030	535	632	11.7	31.77	495	67.8	130	2.9	37.3	37.4	69.1	2.7	5.41
UPN 320	59.5	320	100	14	17.5	17.5	8.75	75.8	246	M22	58	62	0.982	16.5	UPN 320	59.5	10870	679	826	12.1	47.11	597	80.6	152	2.81	43	66.7	96.1	2.6	4.82
UPN 350	60.6	350	100	14	16	16	8	77.3	282	M22	56	62	1.047	17.25	UPN 350	60.6	12840	734	918	12.9	50.84	570	75	143	2.72	40.7	61.2	114	2.4	4.45
UPN 380	63.1	380	102	13.5	16	16	8	80.4	313	M24	59	60	1.11	17.59	UPN 380	63.1	15760	829	1014	14	53.23	615	78.7	148	2.77	40.3	59.1	146	2.38	4.58
UPN 400	71.8	400	110	14	18	18	9	91.5	324	M27	61	62	1.182	16.46	UPN 400	71.8	20350	1020	1240	14.9	58.55	846	102	190	3.04	44	81.6	221	2.65	5.11

Tablo 27. Yarım I ve IPE profilleri



½ I

e_x : x–x ağırlık ekseninin mesâfesi



½ IPE

İşâret	Ölçüler					A	G	Eğilme Eksenleri İçin						e_x
								x - x			y - y			
	h	b	s	t	r_1 r	A	G	J_x	W_x	i_x	J_y	W_y	i_y	e_x
	mm	mm	mm	mm	mm	cm^2	kg/m	cm^4	cm^3	cm	cm^4	cm^3	cm	cm
½ I	DIN 1025 Blatt 1'e göre yarım I profili													
140	70	66	5,7	8,6	5,7	9,15	7,20	37,9	7,26	2,03	17,6	5,35	1,40	1,79
160	80	74	6,3	9,5	6,3	11,4	8,95	62,2	10,5	2,34	27,4	7,40	1,55	2,04
180	90	82	6,9	10,4	6,9	14,0	11,0	99,7	14,9	2,65	40,7	9,90	1,71	2,30
200	100	90	7,5	11,3	7,5	16,7	13,1	144	19,4	2,94	58,2	12,9	1,87	2,56
220	110	98	8,1	12,2	8,1	19,8	15,5	208	25,4	3,24	81,0	16,6	2,02	2,83
240	120	106	8,7	13,1	8,7	23,0	18,1	289	32,5	3,54	110	20,8	2,20	3,09
260	130	113	9,4	14,1	9,4	26,7	20,9	396	41,1	3,85	144	25,4	2,32	3,37
280	140	119	10,1	15,2	10,1	30,5	23,9	528	51,1	4,16	182	30,5	2,45	3,66
300	150	125	10,8	16,2	10,8	34,5	27,1	691	62,6	4,47	225	36,0	2,56	3,96
320	160	131	11,5	17,3	11,5	38,9	30,5	888	75,7	4,78	277	42,3	2,67	4,26
340	170	137	12,2	18,3	12,2	43,3	34,0	1130	90,6	5,10	336	49,1	2,80	4,56
360	180	143	13,0	19,5	13,0	48,5	38,1	1420	108	5,40	408	57,1	2,90	4,86
380	190	149	13,7	20,5	13,7	53,5	42,0	1750	126	5,72	486	65,3	3,02	5,16
400	200	155	14,4	21,6	14,4	58,9	46,2	2140	147	6,02	578	74,6	3,13	5,46
425	212,5	163	15,3	23,0	15,3	66,2	51,9	2710	176	6,40	717	88,0	3,30	5,82
450	225	170	16,2	24,3	16,2	73,4	57,6	3400	209	6,80	861	101	3,43	6,21
475	237,5	178	17,1	25,6	17,1	81,4	63,9	4210	245	7,19	1040	117	3,60	6,58
500	250	185	18,0	27,0	18,0	89,7	70,4	5150	286	7,58	1240	134	3,72	6,96
½ IPE	DIN 1025 Blatt 5'e göre paralel başlıklı yarım I profili													
140	70	73	4,7	6,9	7	8,21	6,45	33,0	6,14	2,01	22,4	6,15	1,65	1,62
160	80	82	5,0	7,4	9	10,0	7,89	52,9	8,57	2,29	34,1	8,34	1,84	1,84
180	90	91	5,3	8,0	9	12,0	9,40	80,3	11,5	2,59	50,4	11,1	2,05	2,05
200	100	100	5,6	8,5	12	14,2	11,2	117	15,1	2,87	71,2	14,2	2,24	2,25
220	110	110	5,9	9,2	12	16,7	13,1	165	19,3	3,15	102	18,6	2,48	2,45
240	120	120	6,2	9,8	15	19,6	15,4	227	24,3	3,41	142	23,7	2,69	2,63
270	135	135	6,6	10,2	15	23,0	18,0	346	32,8	3,88	210	31,1	3,02	2,97
300	150	150	7,1	10,7	15	26,9	21,1	509	43,6	4,35	302	40,3	3,35	3,32
330	165	160	7,5	11,5	18	31,3	24,6	717	55,8	4,78	394	49,3	3,55	3,65
360	180	170	8,0	12,7	18	36,4	28,5	992	70,8	5,22	521	61,3	3,79	3,99
400	200	180	8,6	13,5	21	42,2	33,2	1450	93,7	5,86	659	73,4	3,95	4,52
450	225	190	9,4	14,6	21	49,4	38,8	2220	129	6,70	838	88,4	4,12	5,28
500	250	200	10,2	16,0	21	57,8	45,3	3260	172	7,52	1070	107	4,31	6,01
550	275	210	11,1	17,2	24	67,2	52,8	4670	225	8,33	1330	127	4,45	6,77
600	300	220	12,0	19,0	24	78,0	61,2	6500	288	9,13	1690	154	4,66	7,48